Annie Prasanna Manoharan

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Education

University of Arkansas Aug 2024 – May 2026
Master of Science in Computer Engineering *Fayetteville, AR*
Coursework: Algorithms, Database Management Systems, Advanced DBMS, Data Mining, Artificial Intelligence, Machine Learning, Image Processing, Privacy Enhancing Techniques

Sri Ramakrishna Engineering College Aug 2019 – May 2023
Bachelor of Engineering in Electronics and Communication *India*

Technical Skills

Cloud and Tools: Interaction design, Prompt engineering, CSS, TypeScript, Rapid prototyping, Human-computer interaction, Git, Vector embeddings, HTML
Data Engineering:
Languages: SQL, Python
Machine Learning and AI: Flask, Streamlit, LangChain, RAG, FAISS, Ollama, Llama 3.2, Pandas, Scikit-learn

Projects

Interactive AI Study Partner Prototype *Python, Streamlit, Claude API, FAISS, HTML, CSS*

- Prototyped a conversational study partner that asks guiding questions instead of giving answers, exploring interaction patterns where the AI adapts to what the learner already knows
- Built session logging that records what a learner can do unaided over time, so progress is measured by growing independence rather than time spent in the tool
- Iterated on the interface through informal user sessions, adjusting prompts, pacing, and feedback based on where learners got stuck

Local LLM-Powered News Research Platform *Python, LangChain, Streamlit, FAISS, Ollama, Llama 3.2*

- Built an end-to-end local ETL pipeline using Streamlit and WebBaseLoader to ingest, clean, chunk, and preprocess unstructured web content on demand
- Designed a local vector store with FAISS and nomic-embed-text to serialize high-dimensional embeddings, bypassing redundant ingestion and reducing processing latency
- Implemented a RAG pipeline via LangChain orchestrating semantic queries against vectorized data using local Llama 3.2 instances

Professional Experience

Data Engineering Intern Jul 2024 – Present
InsOps Inc. *Chicago, Illinois*

- Developed data pipelines in AWS using AWS Glue, Lambda, EMR, SQL, and Apache Airflow for orchestration, with S3 cloud storage, for scalable data ingestion, transformation, validation, and analytics
- Built data workflows integrating SQL Server, Azure SQL, AWS S3, and cloud databases, preparing structured datasets for downstream data science and analytics

Teaching Assistant Sep 2024 – May 2026
University of Arkansas, Dept. of Computer Science and Engineering *Fayetteville, AR*

- Supported 100+ students in Programming Foundations through laboratory instruction, debugging assistance, and mentorship in C++ programming fundamentals

Research Assistant Sep 2024 – Jan 2026
AICV Lab, University of Arkansas *Fayetteville, AR*

- Built reusable preprocessing workflows to segment, organize, and transform raw image/video data into structured inputs for downstream processing and evaluation
- Evaluated results using mAP, MOTA, IDF1, PSNR, SSIM, and FID

Project Engineer Jun 2023 – Jun 2024
Soliton Technologies *India*

- Designed and developed a C# WPF desktop application integrating C#, Python, and LabVIEW-based test workflows to streamline application usage and improve test automation
- Developed customized software solutions for Dräger as the sole developer, automating approximately 94% of manual test cases while meeting medical-device quality and validation requirements